14trees.org

14 TREES FOUNDATION
Ecological Report

BIODIVERSITY & RESEARCH GROUP REPORT

FY 2024-25

EXECUTIVE SUMMARY

Biodiversity is a crucial indicator of ecosystem recovery, as biodiverse systems support diverse plants and animals that can coexist long-term. This coexistence is vital for climate-change adaptation and mitigation. At 14 Trees Foundation (14 TF), we are dedicated to a holistic approach to ecosystem revival, with biodiversity enhancement as a primary focus.

14 TF employs Assisted Natural Regeneration (ANR) to reforest barren lands dealing with pressure from over-grazing and intensive burning. Through ANR, we assist and protect plants until they can sustain themselves. This involves creating a supportive ecosystem with healthy soil, adequate water, and optimal microclimatic conditions. Consequently, our work extends beyond simply planting native flora to encompass all aspects of ecosystem recovery, including improving soil health, recharging aquifers, and providing nature-based sustainable livelihoods. The Biodiversity team within 14 TF is responsible for quantifying and documenting the changes in biodiversity as our restoration work progresses. Additionally, the team designs and executes research projects that align with the broader field of Ecological Sciences, partnering with academic institutions, research organizations, and consulting firms for all our projects. This report summarizes the Biodiversity team's activities in FY 2024-25 and outlines our roadmap for FY 2025-26.

CORE TEAM

Sanjeev Jagtap

Sanjeev Naik

Yogita Karpate

TECHNICAL & ACADEMIC PARTNERS

Revati Gindi

Pratik Purohit

Monali Shah

Ashwini Bhilare

IISER-Pune

Ecological Society-Pune

This report is developed in collaboration with our technical partners, Ecological Society Pune (ES) and Eco Connect (EC). We partnered with ES for the biodiversity survey in the fiundation land in Vetale village, Khed taluk, and with EC for baseline surveys in the Gairans. We deeply appreciate the mentorship we receive from our technical partners' and financial support from our funding partners.

261

Flora species recorded in foundation land

112

Fauna species recorded in foundation land

12

Gairans surveyed

14trees.org

TABLE OF CONTENTS

14trees.org

BIODIVERSITY SURVEY METHODOLOGY AND RESULTS

2020-2021

Since FY 2020-21, our team has conducted periodic surveys to document the flora and fauna at our sites. The primary goal of these surveys is to record the species richness (number of species) of various plant (floral taxa) and animal groups (faunal taxa). Rapid Field Assessments, as it is called, is apt for collecting baseline information for the species thriving in the area.

Our methodology over the years

2023-2024

From FY 2023-24, we adopted a more systematic assessment methodology (quadrat based data collection). We repeated survey of the same quadrats in every season to understand a seasonal trend (More details in the next section)

2024-2025

We continued with the same methodology in FY 2024-2025.

2025-2026

To gain deeper insights into the functioning of nature, we plan to document additional parameters related to soil, water, and other ecologically significant variables. Recognizing the interdisciplinary nature of this work and the need to scale up our activities with technology, we are collaborating with various organizations. and experts.

Our survey methodology evolved over years. We started with ***Rapid Field Assessments (RFA)***, following random trails on the site. The main intention was to have maximum coverage of the land, all habitats, and terrains. Although ***RFA*** helped us understand what is thriving in our area, there is always a possibility of missing out on some important species. Therefore, from FY 2023-24, we decided to switch to **quadrat-based** assessment, in which we established **23** permanent plots of **10 m X 10 m** size for documenting flora. The placement of the plots or quadrats was done in such a way as to cover all topographies and vegetation types. The location of the quadrats is marked in blue in Fig 1.

Figure 1: Point locations of quadrats

Details of Survey Methodology in FY 2024–25

For fauna, we used Point counts and Active Searching method in each quadrat and its 10-meter buffer, with each observation lasting 30–40 minutes. We also took into account indirect evidences such as pugmarks, scats, and animal droppings while logging the data. During the surveys, we also noted abiotic features such as soil substrate and climate for the microhabitat.

Flora

Our site is dominated by herbaceous plants, followed by grasses, trees and other plant groups. A large portion of our land is still in early stages of restoration and therefore colonized by herbs more than any other plant groups. This pattern is expected to shift towards dominance by trees and shrubs as the site mature.

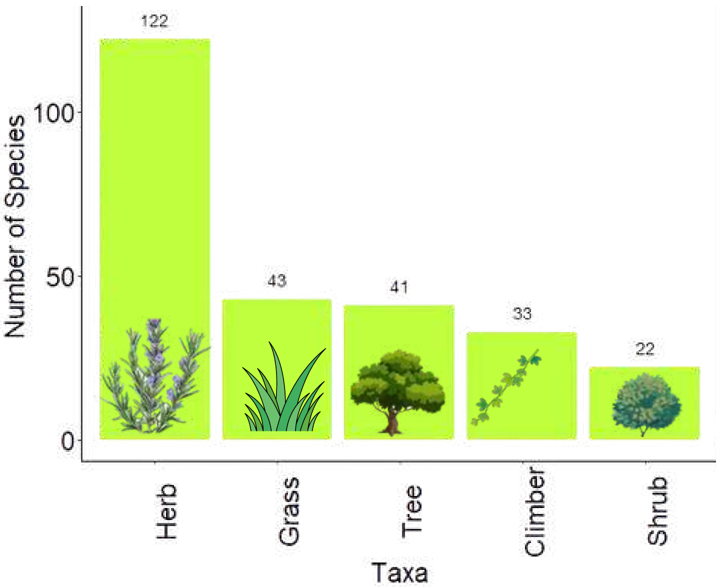


Figure 2: Number of species from different plant groups

RESULTS

We observed varying proportions of plant groups—trees, grasses, herbs, climbers, and shrubs—across different zones. These variations may be attributed to factors such as topography, microhabitat, and the duration since restoration efforts were initiated (Fig 2)

14trees.org

The zone-vise analysis indicates that herbaceous species are still the most common across all areas. However, Zone 1, being the most developed, shows a growing number of species from other plant groups. As the site continues to mature, we anticipate that all plant groups will be equally represented, leading to high species richness and evenness, a key indicator of a thriving ecosystem (Fig 3)

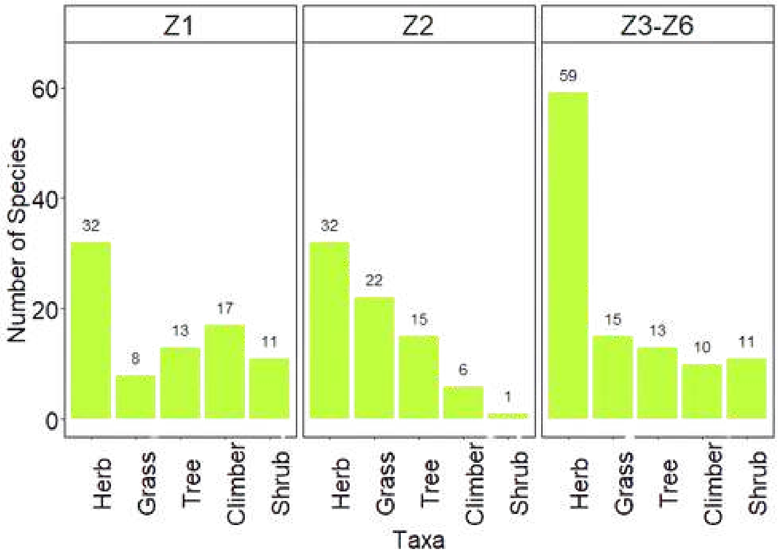


Figure 3: Zonal trend

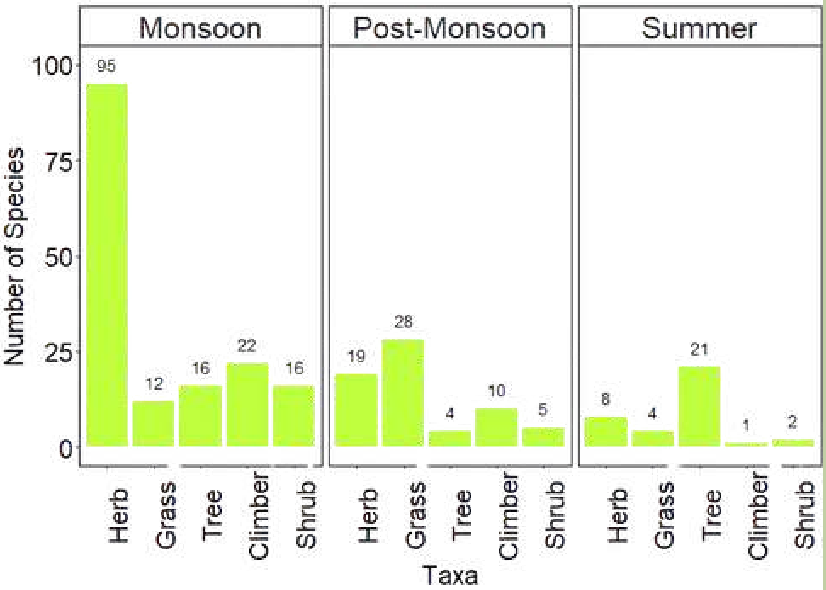


Figure 4: Seasonal trend

During the monsoon season, herbaceous species are most common, with grasses becoming more prevalent after the monsoon. However, their presence decreases in the summer, when records predominantly indicate the presence of trees. This is because grasses and most of the shrubs are annuals, meaning their life cycles end in one year whereas trees and woody shrubs stay rooted at their place throughout the year. The grass height post-monsoon shoots up, sometimes making it difficult to enter the plots we marked for observations. This is evident by the low number of trees observed post-monsoon (Fig 4)

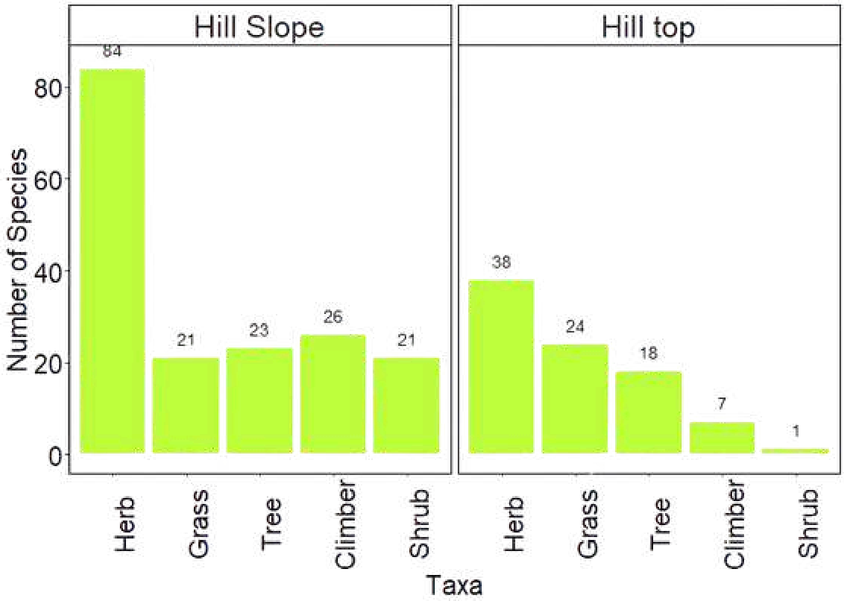


Figure 5: Topography vise trend

Hilltops are predominantly populated by herbaceous and grass species. This is likely because the barren plateaus have minimal soil, which only supports the growth of herbs and grasses. Conversely, hill slopes exhibit a more balanced representation of species, potentially due to a favorable microenvironment that supports a wider variety of plant groups (Fig 5).

To pinpoint the exact reasons behind pattern in Fig 5, we should investigate soil and microclimatic parameters in more detail.

Our restoration site at Vetale has a wide presence of open-scrubland with some forest that has come up due to our restoration work progressed Finally, the habitat-vise distribution indicates a dominance of herbaceous species in open scrub areas, a characteristic feature of scrublands. Conversely, the forests exhibit greater evenness, though herbs still dominate the understory (Fig 6)

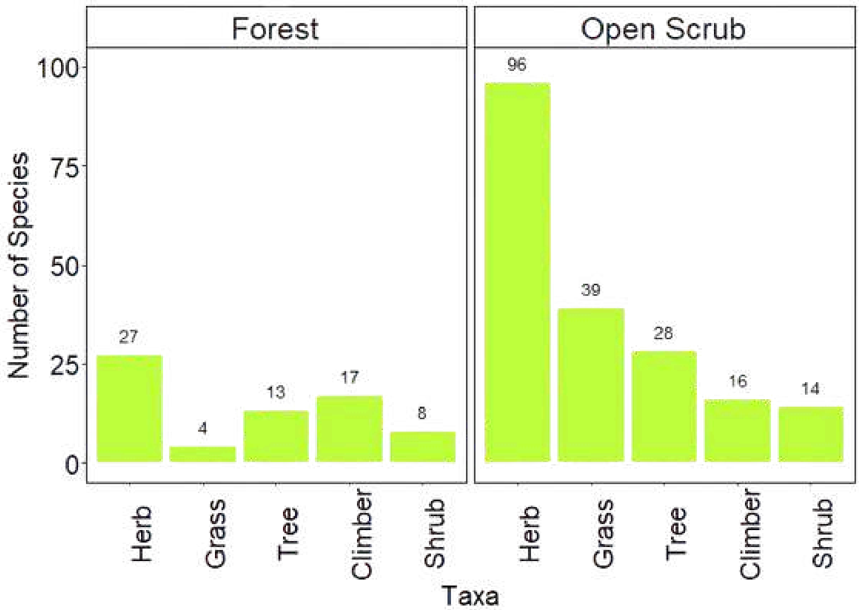


Figure 6: Habitat vise trend

14trees.org

Fauna

During our site assessment, we observed birds and insects, which are often the first to colonize restored areas and indicate ecosystem recovery. As the ecosystem develops, we anticipate greater diversity within these groups and the introduction of other faunal groups, such as mammals. We also identified several reptile species and a few amphibians in certain areas.

RESULTS

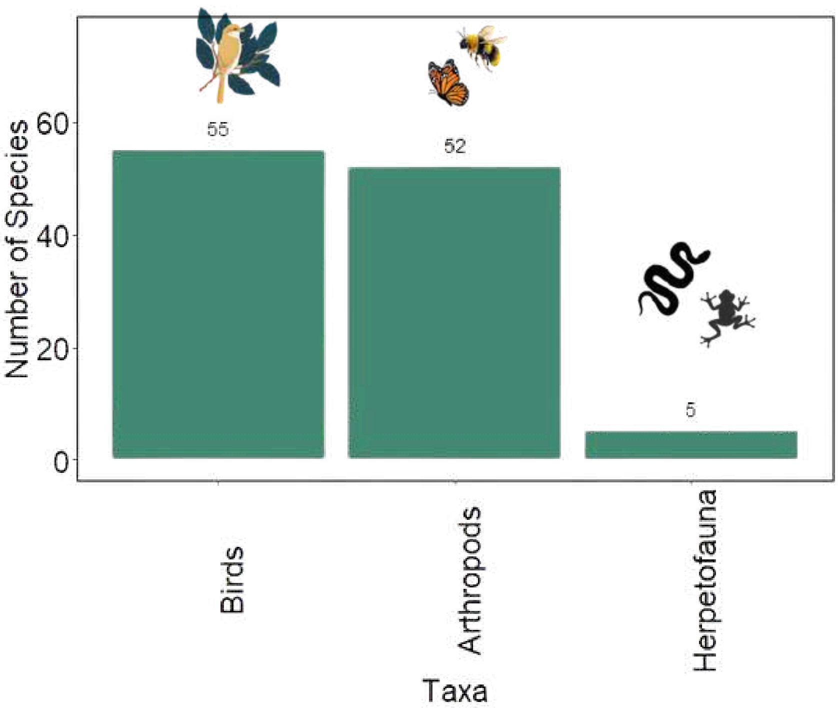
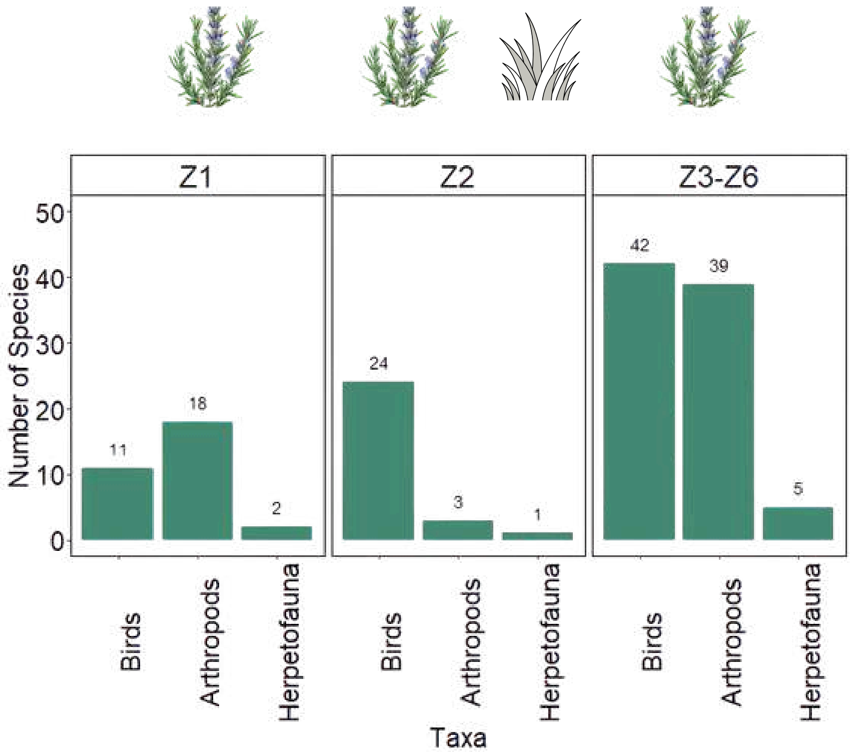


Figure 7: Overall species trend

The observed faunal taxa variations (birds, arthropods, and herpetofauna) across different zones may be related to vegetation, food and shelter availability, and other factors. The increasing presence of fauna with diverse feeding habits indicates improving ecosystem health.
Note that these results are based on observed species during our field survey, and some species may have been missed (Fig 7)

.

14trees.org



Insect and bird patterns differ across zones due to environmental factors like vegetation, topography, and season. Zones 3-6, now under restoration, host more grass-dependent insects than Zone 1's mature forest, which has less insect and bird diversity. Zones 3-6 may have more species due to their size relative to Zone 1. As reforestation progresses, we anticipate an increase in species (Fig 8)

Figure 8: Zonal trend

Surveys indicate more bird and insect species on hill slopes than on hilltops. This difference correlates with vegetation: hill slopes are mainly herbaceous with diverse flora, while hilltops have fewer tree and herb species, reducing insect and bird diversity (Fig 9).

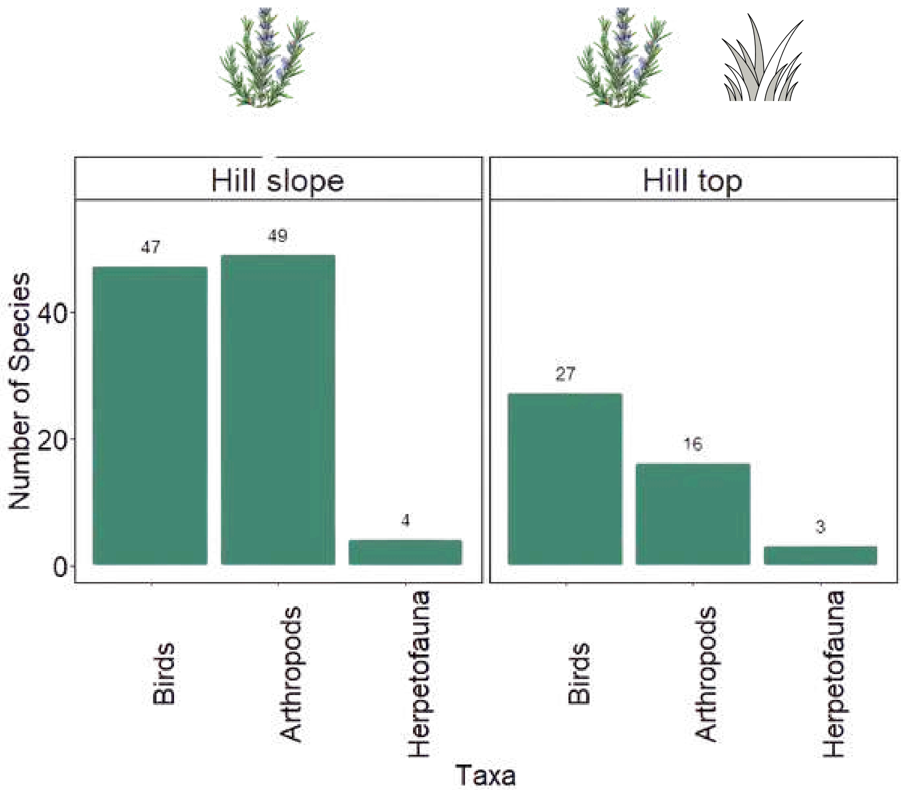


Figure 9: Topography vise trend

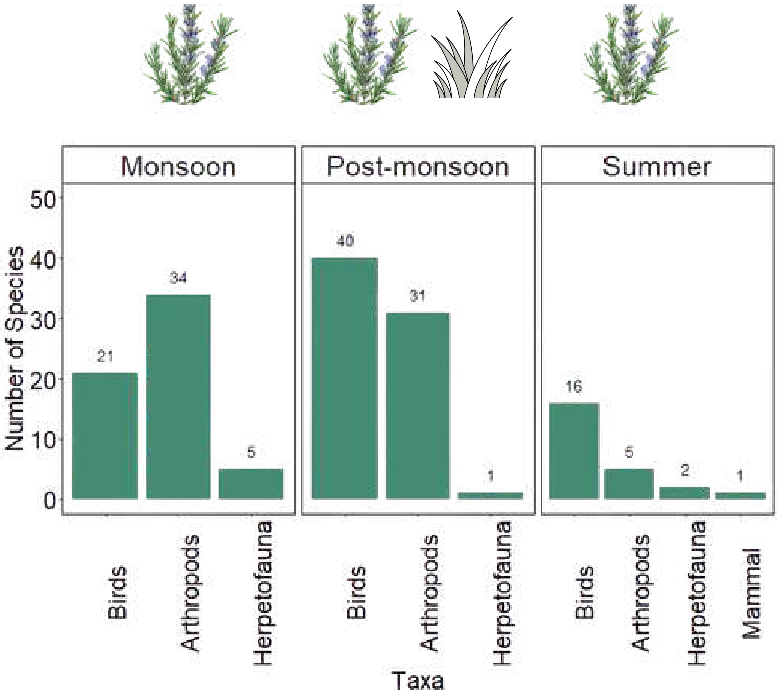


Figure 10: Seasonal trend

During monsoon and post-monsoon, the number of observed species increased, likely due to greater prey and food availability. Insect species peaked in the monsoon, and bird species peaked post-monsoon. Tree immaturity may limit summer habitats for faunal groups, given the plantation's initial phase (Fig 10)

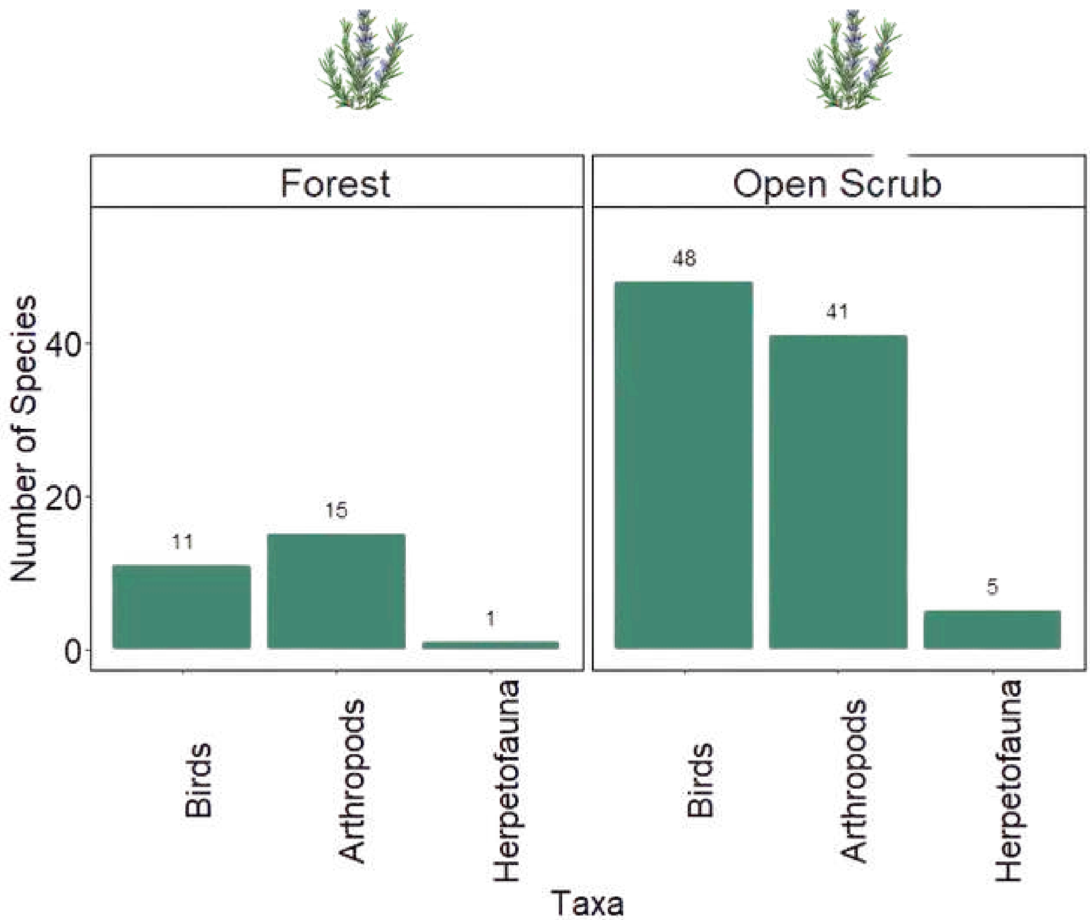


Figure 11: Habitat vise trend

The major portion of the restoration site still is open-scrubland, consequently resulting in more species observed in open-scrub areas (Fig 11).

GAIRAN BIODIVERSITY SURVEY

LOCATIONS

In FY 2024-25, we completed a survey of 13 Gairans (community grazing lands), where we recorded observations of flora and fauna. The locations of these gairans are shown with green markers in Fig 12. Most of these Gairans are located within the Khed taluka, except a few in the neighbouring talukas. These Gairans have been undergoing overgrazing pressure for a long time, leading to severe degradation of land, which is evident by establishment of only a few plant species and corresponding fauna. We have started restoration work in some of the gairans, however, restoration in most of these sites is in its initial phase. The biodiversity surveys in these locations were intended to record the status before we begin restoration work (baseline data). This data will be useful for comparative analysis at different stages of restoration.

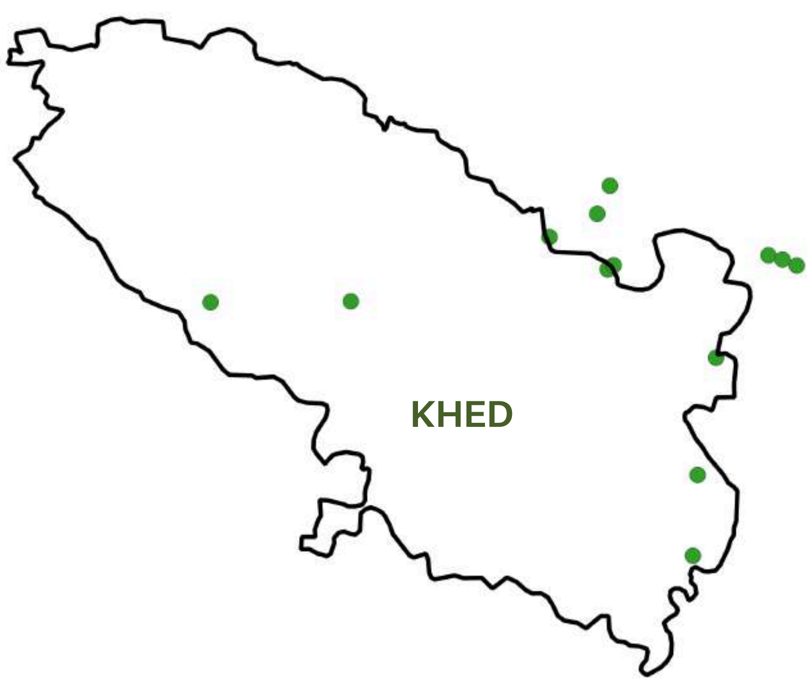


Figure 12: Gairan locations

FLORA

GAIRAN NAME	HERB SPECIES	GRASS SPECIES	TREES SPECIES	SHRUB SPECIES	CLIMBER SPECIES
Avsari	26	19	9	3	2
Chinchoshi	21	11	10	9	4
Devtoranne	22	12	12	4	1
Dhamani	41	12	12	5	5
Ghargotewadi	28	26	19	8	6
Loni	31	15	10	5	2
Pabal	16	15	9	4	4
Pargaon-1	51	29	16	6	1
Pargaon -2	25	17	3	1	3
Peth	44	NA	8	3	NA
Pur	41	20	13	9	7
Siddhegavan	42	16	5	4	3
Wafgaon	32	10	15	3	4

FAUNA

GAIRAN NAME	ARTHROPODS	BIRDS
Avsari	5	17
Chinchoshi	3	32
Devtoranne	11	17
Dhamani	10	23
Ghargotewadi	4	14
Loni	9	22
Pabal	7	25
Pargaon-1	29	25
Pargaon-2	16	7
Peth	23	33
Siddhegavan	19	18
Pur	35	26
Wafgaon	7	20


24/12/2024
24/12/2024
28/10/2024
24/09/2024
24/09/2024
24/09/2024

MOTH SPECIES SURVEY

14trees.org

Moths have a wide range of important ecological roles, from pollinating nocturnal flowers and serving as a crucial food source for various animals to participating in the decomposition of organic material. Furthermore, certain moth species are susceptible to environmental changes like pollution, habitat loss, and climate change, making them potentially valuable indicators of environmental health.

Given their ecological significance, we've begun surveying moth diversity within our foundation land. This work is happening in collaboration with IISER-Pune. Our initial efforts involve observations made on new moon nights. So far, we've covered six locations within the foundation land. These sites include grassland patches, plantations, areas near human habitation, and the edges of agricultural land and plantations.

METHODS

We set up a white screen with UV lights arranged along the border of the frame. UV lights work as trap for Moth species. We make observations every two hours from 8 pm to 12 am.

RESEARCH PROJECT

14trees.org

BACKGROUND

In FY 2024-25, we proposed a research project in partnership with RM Tulpule Charitable Trust, TIFR-NCBS-Bengaluru, and Ecological Society-Pune. In this project we aim to conduct a comparative analysis of traditional land-use (grazing and burning) with restoration on ecological functions such as aboveground carbon storage, soil characteristics, plant diversity, faunal diversity (mainly avian diversity). The proposed duration for this study is 3 years.

METHODS

We identified a few locations within the foundation land that included,
1) Grasslands where grazing and burning is allowed
2) Grasslands where only grazing takes place &
3) Planted patches.
We marked a few quadrats in these locations where we will do seasonal sampling for different parameters related to carbon sequestration, soil characteristics and species diversity.

PROGRESS

We finished collecting baseline data for species diversity and soil sampling along with water sampling from few ponds. These samples will be processed for chemical and physical properties, along with microbial counts. Our next focus will be on collecting on-ground data on above ground carbon stocks for all the land uses.

FOCUS AREAS FOR FY 2025-26

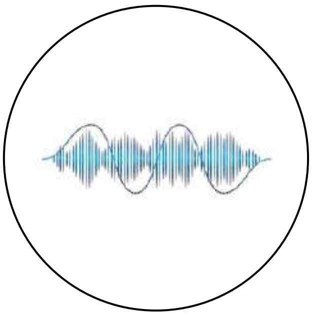
Going forward in FY 2025-26, we would like to expand our focus into several other dimensions of natural ecosystem. Some of the potential areas that we would like to explore are listed in the following section. We look forward to partner with experts in these fields to execute pilot projects in the upcoming quarters of FY 2025-26.

Enhancing Biodiversity surveys
Collection of wider range of ecological parameters related to species and habitat

Baseline data collection for newly acquired Gairans
Explore new gairans for baseline data acquisition

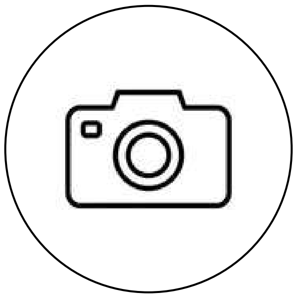
Systematic Surveys in Gairans
Detailed ecological surevys of gairans for which baseline is available

Insect diversity
Systematic survey of insects



Bioacoustics
Acoustics survey of bird calls

TGBS standards
Aligning socio-ecological monitoring methods with The Global Biodiversity Standards



Camera trapping
Camera trapping for mammals

Socio-economic data
Documenting socio-economic parameters related to restoration work

Soil enrichment & Biochar experiments
Impact of restoration on soil health & potential of biochar to improve soil quality

Remote Sensing & GIS
Spatial analysis of land-use and habitat characteristics

We would like to express our sincere gratitude to the core group members at 14 Trees Foundation, collaborators, funding partners, and the on-site staff for their continued support and encouragement. We hope that we will continue working collaboratively towards our common goal of restoring barren lands to their natural habitats with even more enthusiasm and rigor than before.

Site Address
14 Trees Foundation
Vetale, Taluka Khed
Pune - 412404
Maharashtra

Pune Office
14 Trees Foundation
Atal Incubation Centre (AIC)
IISER-Pune (411008)
Maharashtra

Email Addresses
General inquiries: contact@14trees.org
Partnerships and Corporate: csr@14trees.org
Research & biodiversity group: yogita.karpate@14trees.org

14trees.org

GLOSSARY

1. Foundation land ....Land acquired by 14 Trees Foundation in Vetale for restoration work
2. Gairan..Community grazing land
3. Zone..Divisions of foundation land for restoration planning
4. Flora..Plants
5. Shrubs..Small trees
6. Climbers..Climbing plants
7. Fauna..Animals & insects
8. Herpetofauna..Amphibians & reptiles
9. Arthropods...Insects
10. Bioacoustics...Study methodology that focuses on analyzing animal communication
11. Camera-trapping..Study methodology involving recording videos and images of animal movement using motion sensors.
12. Remote sensing...Research methodology including study of satellite data, drone imagery etc.